



K L Deemed to be University
CSE-1 -- KLVZA
Course Handout
2025-2026, Odd Sem

Course Title	:INDIAN ANCIENT KNOWLEDGE SYSTEMS: VEDIC MATHEMATICS
Course Code	:25UC0017
L-T-P-S Structure	: 2-0-0-0
Pre-requisite	:
Credits	: 0
Course Coordinator	:Bhargavi Yella
Team of Instructors	:
Teaching Associates	:

Syllabus : Introduction to Indian knowledge system, Indian mathematicians and their contribution to the world, Vedic Mathematics: origin of Vedic mathematic, 16 sutras of Vedic Mathematics by swami Bharti Krishna Tirthji Maharaj Basic arithmetic operation by Vedic sutras: Ekadhikena Poorvena, Ekanyunena Poorvena, Nikhilam Navatacharamam Dasatah, Multiplication by Urdhva-tiryagbhyam, Sunyam Samya Samuchaye, Division: Paravartya Yojayet and Nikhilam Method, Anurupyena, Yavadunam, Addition by Shubh method. Numbers, Ratio, Proportion, Variation, Averages, Percentages, Profit & loss, Time & Distance, Time & Work and related Shakuntala Devi Puzzles. Clocks, Calendars, Cubes, logical connectives, Deductions, Linear and circular arrangements, Number and letter series, Coding and Decoding, Binary Logic.

Text Books : 1. Vedic Mathematics or Sixteen simple Mathematical Formulae from the Vedas, SWAMI SRI BHARATI KRSNA TIRTHAJI MAHARAJA, Motilal Banarasi Das Press, MATHA, PURI. 2. R. S. Agarwal, A Modern Approach to Verbal and Non-verbal Reasoning, S Chand Publications, 2018, New Delhi, India. 3. Puzzles to Puzzles by Sakunthaladevi

Reference Books : 1. Vedic Mathematics for Speed Arithmetic, M Satyamoorthni Hebbar, Vasan Publications, Bangalore. 2. Vedic Mathematics, Dr. Florentin Smarandache, Dr. W. B. Vasantha Kandasamy

Web Links : 1. <https://iiva.in/vedic-maths-course/online/>
 2. <https://www.youtube.com/watch?v=p2WankcGP3Q&list=PLbMVogVj5nJThf31TNSQzuN7zqxe7HdRN>

MOOCS : https://www.udemy.com/course/vedicmaths/?--=&utm_source=adwords&utm_medium=udemyads&utm_campaign=LongTail_la.EN_cc.INDIA&utm_content=deal4584&utm_term=_.ag_118445032537_.ad_618853564450_.kw_.de_c_.dm_.pl_.ti_aud-1738475842996%3Aadsa-1212271230479_.li_9301694_.pd_.&matchtype=&gclid=Cj0KCQjw1_SkBhDwARIsANbGpFt73yGfw_ZEjkitrMSCncJVSpI_hk0bf0xgAdlyc2svm9-a4H4W-OAaAnFtEALw_wcB

Course Rationale : India has a rich legacy of intellectual and scientific advancement, with mathematics being a cornerstone of its ancient knowledge systems. This course introduces students to the Indian Knowledge System (IKS) with a focus on the mathematical innovations of ancient Indian scholars and the powerful techniques of Vedic Mathematics as rediscovered by Swami Bharati Krishna Tirthaji Maharaj. Vedic Mathematics is a collection of 16 sutras (formulae) and sub-sutras that simplify arithmetic operations and algebraic manipulations. These methods promote faster, more accurate calculations and enhance mental math abilities. Alongside, the course includes modern quantitative topics such as numbers, ratio, proportion, percentages, time and work, and logical reasoning, enriched with puzzles inspired by Shakuntala Devi, known as the "Human Computer". This integration of ancient wisdom with contemporary problem-solving aims to develop critical thinking, mental agility, and appreciation for India's scientific heritage, making learners competent for both academic pursuits and competitive exams.

Course Objectives : To solve quantitative problems using both traditional and Vedic methods in areas such as numbers, percentages, ratios, and time-based calculations. To develop problem-solving strategies using logical reasoning and analytical thinking and appreciate the synergy between traditional Indian mathematics and modern mathematical education. Cultivate speed, efficiency, and confidence in mathematical thinking through practice with puzzles and real-world problems.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Interpret the origin and philosophy of Vedic Mathematics as rediscovered and structured by Swami Bharati Krishna Tirthaji Maharaj and Summarize the purpose and meaning of the 16 Vedic Mathematics sutras, including their mathematical significance and application areas.	PSO1,PO1	2
CO2	Apply Vedic Mathematics sutras to perform efficient addition, subtraction, multiplication, and division operations.	PSO1,PO1	3
CO3	Apply logical reasoning principles and problem-solving strategies to solve related Shakuntala Devi Puzzles.	PSO1,PO1	3
CO4	Apply reasoning concepts to solve problems involving clocks, calendars, cubes, deductions, logical connectives, and arrangements.	PSO1,PO1	3

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3
CO1	2	Btl-1 Recall key concepts of Indian philosophy, contributions of Indian mathematicians, and their influence on global mathematical development.	Btl-2 Describe the interdisciplinary relationship between Indian philosophy and mathematics, and explain the significant contributions of Indian mathematicians to the evolution of global mathematical knowledge.	
CO2	3	Btl-1 List and recall the Vedic Mathematics sutras used for addition, subtraction, multiplication, and division.	Btl-2 Explain the use of Vedic Mathematics sutras in performing addition, subtraction, multiplication, and division operations.	Btl-3 Solve addition, subtraction, multiplication, and division problems using appropriate Vedic Mathematics sutras.
CO3	3	Btl-1 Recall basic logical reasoning principles and problem-solving techniques used in Shakuntala Devi's mathematical puzzles.	Btl-2 Explain the logical reasoning principles and problem-solving strategies used to solve puzzles inspired by Shakuntala Devi.	Btl-3 Apply logical reasoning principles and problem-solving strategies to solve mathematical puzzles inspired by Shakuntala Devi.

CO4	3	Btl-1 Recall fundamental concepts and basic rules related to reasoning problems on clocks, calendars, cubes, deductions, logical connectives, and arrangements	Btl-2 Explain the concepts and reasoning methods used to solve problems related to clocks, calendars, cubes, deductions, logical connectives, and arrangements.	Btl-3 Apply reasoning concepts to solve problems involving clocks, calendars, cubes, deductions, logical connectives, and arrangements.
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PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PSO1	An ability to design and develop software projects as well as Analyze and test user requirements.
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-1	Introduction to Course Handout	NA	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
2	CO1	COI-1	Introduction to Indian knowledge system	TB-1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
3	CO1	COI-2	Indian mathematicians and their contribution to the world	TB-1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
4	CO1	COI-2	Origin of Vedic mathematics	TB-1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
5	CO1	COI-2	16 sutras of Vedic Mathematics by swami Bharti Krishna Tirthji Maharaj	TB-1	Chalk,Talk	ALM,End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
6	CO2	COI-2	Basic arithmetic operation by Vedic sutras: Ekadhikena Poorvena, Ekanyunena Poorvena	TB-1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
7	CO2	COI-2	Basic arithmetic operation by Vedic sutras: Nikhilam Navatacharamam Dasatah, Multiplication by Urdhva-tiryagbhyam	TB-1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
8	CO2	COI-3	Basic arithmetic operation by Vedic sutras: Paravartya Yojayet	TB-1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
9	CO2	COI-3	Basic arithmetic operation by Vedic sutras: Nikhilam Method	TB-1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
10	CO2	COI-3	Basic arithmetic operation by Vedic sutras: Anurupyena, Yavadunam and Addition by Shubh method	TB-1	Chalk,Talk	ALM,End Semester Exam Online,Home Assignment,Semester in Exam-I (MCQ)
11	CO3	COI-2	Number System	R3, Ch1, P12	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
12	CO3	COI-2	Ratio, Proportion and Variation	TB 3, Ch2, P34	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
13	CO3	COI-3	Averages and Percentages	TB4, Ch3, ch4, ch6	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
14	CO3	COI-3	Profit and Loss	TB4, Ch 7	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
15	CO3	COI-	Time and Distance	TB3 AND	Chalk,Talk	End Semester Exam

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
		2		TB4		Online,Home Assignment,Semester in Exam-II (MCQ)
16	CO3	COI-3	Time & Work	TB3, TB4,	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
17	CO3	COI-3	Time & Work	TB3, TB4	Chalk,Talk	ALM,End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
18	CO4	COI-2	Clocks	TB3, TB4	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
19	CO4	COI-2	Calendars	TB3, RF1	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
20	CO4	COI-3	Cubes	TB4, RF2	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
21	CO4	COI-2	logical connectives	TB4, RF2	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
22	CO4	COI-2	Deductions	TB4, RF3	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
23	CO4	COI-3	Linear and circular arrangements	TB4, RB2	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
24	CO4	COI-3	Binary logic	TB4, RB2	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
25	CO4	COI-3	Number and letter series	TB4, RB2	Chalk,Talk	End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)
26	CO4	COI-3	Coding and Decoding	TB4, RB2	Chalk,Talk	ALM,End Semester Exam Online,Home Assignment,Semester in Exam-II (MCQ)

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Remember the course rationale and evaluation methods

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Briefing on Course Rationale	1	Talk	--- NOT APPLICABLE ---
20	Briefing on Course Handout	1	Talk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Recall the significance and scope of Indian Knowledge Systems in shaping global scientific and mathematical thinking

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Importance and relevance of Indian knowledge system	1	Talk	--- NOT APPLICABLE ---

20	Indian knowledge system in education, science, and math	1	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3**Session Outcome: 1** Identify the key contributions of prominent Indian mathematicians

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	contributions of prominent Indian mathematicians such as Aryabhata, Bhaskara, Brahmagupta, and Srinivasa Ramanujan	2	Chalk	--- NOT APPLICABLE ---
20	how these contributions influenced global mathematics	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4**Session Outcome: 1** Explain the historical background and origin of Vedic Mathematics.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explain the historical origin of Vedic Mathematics.	2	Chalk	--- NOT APPLICABLE ---
20	Describe the contribution of Swami Bharati Krishna Tirthaji.	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5**Session Outcome: 1** Describe and interpret the structure of Vedic Mathematics – the 16 Sutras

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	One minute paper
20	Overview of the 16 Sutras	2	Chalk	Quiz/Test Questions
20	Display each sutra with its literal translation and a simple interpretation.	2	Chalk	Quiz/Test Questions
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 1 explain the concept of Ekadhikena Poorvena and interpret its use in squaring numbers ending in 5

Session Outcome: 2 describe the concept of Ekanyunena Poorvena and explain its use in multiplication involving numbers near base values.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Ekadhikena Poorvena-Squaring numbers ending in 5.	2	Chalk	--- NOT APPLICABLE ---
20	Ekanyunena Poorvena	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 1 Explain the multiplication of any numbers

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Nikhilam Navatacharamam Dasatah	2	Chalk	--- NOT APPLICABLE ---

20	Urdhva-tiryagbhyam	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Apply it in basic arithmetic division where traditional methods are lengthy

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Paravartya Yojayet	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Apply it in basic arithmetic division where traditional methods are lengthy

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Nikhilam Method	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Apply these sutras to simplify squaring and multiplication of numbers that are not directly near 10, 100, etc., using flexible base selection and proportional adjustment.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---

20	Anurupyena and Yavadunam	3	Chalk	One minute paper
20	Addition by Shubh method	3	Chalk	One minute paper
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Apply logical reasoning principles and problem-solving strategies to solve related Shakuntala Devi Puzzles.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction of Number System	2	Chalk	--- NOT APPLICABLE ---
20	Problem based on Number system and related puzzles	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Apply the concepts of Ratio, Proportion and Variation to solve related Shakuntala Devi Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concept of Ratio, Proportion and Variation	2	Chalk	--- NOT APPLICABLE ---
20	Problem based on Ratio, Proportion and Variation with Puzzles	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 1 Apply the concepts of Averages and Percentages to solve related Shakuntala Devi Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Averages and Percentages	2	Chalk	--- NOT APPLICABLE ---
20	Problem based on Averages and Percentages with Puzzles	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 1 Apply the concepts of Profit and Loss to solve related Shakuntala Devi Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concept of Profit and Loss	2	Chalk	--- NOT APPLICABLE ---
20	Problem based on profit and loss with Puzzles	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 1 Apply the concepts of Time and Distance to solve related Shakuntala Devi Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction and Basic concept of Time and Distance	2	Chalk	--- NOT APPLICABLE ---

20	Problems based on Time and Distance with Shakuntla Devi Puzzles	3	Chalk	--- NOT APPLICABLE ---
10	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16**Session Outcome: 1** Apply the concepts of Time & Work to solve related Shakuntala Devi Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Chalk	--- NOT APPLICABLE ---
20	Concept of Time & Work	2	Chalk	--- NOT APPLICABLE ---
20	Problems based on Time & Work with puzzles	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17**Session Outcome: 1** Apply the concepts of Time and Work to solve related Shakuntala Devi Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Problems based on Time & Work with puzzles	3	Chalk	--- NOT APPLICABLE ---
20	ALM	3	Talk	Immediate feedback
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18**Session Outcome: 1** Understand the concepts of Clocks to solve related Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	concept of Clocks	2	Chalk	--- NOT APPLICABLE ---
20	Problems based on Clocks with Puzzles	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19**Session Outcome: 1** Understand the concept Calendar of to solve related Shakuntala Devi Puzzles

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	concept of Calendars	2	Chalk	--- NOT APPLICABLE ---
20	Problems based on Calendars with puzzles	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20**Session Outcome: 1** Apply the concepts of cubes to solve the problems

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explain the concept of Cubes	2	Chalk	--- NOT APPLICABLE ---
20	Problems based on Cubes	3	Chalk	--- NOT APPLICABLE

5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21**Session Outcome: 1** Understand the concept of logical Connectives

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduce the concept of logical connectives	2	Chalk	--- NOT APPLICABLE ---
20	Problems based on logical connectives	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22**Session Outcome: 1** Understand the concept of Deductions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction of Deductions	2	Chalk	--- NOT APPLICABLE ---
20	Problems based on Deductions	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23**Session Outcome: 1** Apply the concept of reasoning to solve problems based on Arrangements

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	concept of Arrangements	2	Chalk	--- NOT APPLICABLE ---
20	Problems based on Arrangements	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24**Session Outcome: 1** Apply the concept of reasoning to solve problems based on Binary Logic

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	concept of Binary Logic	2	Chalk	--- NOT APPLICABLE ---
20	Problems on Binary Logic	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25**Session Outcome: 1** Apply the concept of reasoning to solve problems based on Number and letter series

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concept of Number and letter series	2	Chalk	--- NOT APPLICABLE ---
20	Problems on Number and letter series	3	Chalk	--- NOT APPLICABLE ---

5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 26**Session Outcome: 1** Apply the concept of reasoning to solve problems based on Coding and Decoding

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concept & Problems on Coding and Decoding	2	Chalk	--- NOT APPLICABLE ---
20	ALM	3	Talk	Leading question
5	Summary & Conclusion	1	Chalk	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan: NO Delivery Plan Exists**Tutorial Session wise Teaching – Learning Plan**

No Session Plans Exists

Practical Course DELIVERY Plan: NO Delivery Plan Exists**Practical Session wise Teaching – Learning Plan**

No Session Plans Exists

Skilling Course DELIVERY Plan: NO Delivery Plan Exists**Skilling Session wise Teaching – Learning Plan**

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
3	Weekly Homework Assignments	1	Introduction to Indian knowledge system, Indian mathematicians and their contribution to the world, Vedic Mathematics: origin of Vedic mathematic, 16 sutras of Vedic	1. Characterize the Indian knowledge tradition. 2. Demonstrate the discovery of number system and the role of Indian Mathematician. 3.	CO1

			Mathematics by swami Bharti Krishna Tirthji Maharaj	Describe the 16 sutras of Vedic Mathematics and their application in counting.	
5	Weekly Homework Assignments	2	Basic arithmetic operation by Vedic sutras: Ekadhikena Poorvena, Ekanyunena Poorvena, Nikhilam Navatacharamam Dasatah, Multiplication by Urdhva-tiryagbhyam, Sunyam Samya Samuchaye, Divison: Paravartya Yojayet and Nikhilam Method, Anurupyena, Yavadunam, Addition by Shubh method	1. Divide the following by using suitable sutra of Vedic mathematics (a) $10017 \div 89$ (b) $329477 \div 11202$ 2. Calculate SQUARE VALUE OF following numbers by Base Method (a) 2587 (b) 3214 3. Multiply the following by suitable sutra of Vedic Mathematics. (a) Evaluate $37 * 49$ (b) Evaluate $123 * 567$ (c) Evaluate $874 * 693$ (d) Evaluate $2791 * 1284$ (e) Evaluate $64 * 86$ (f) Evaluate $7543 * 8312$ 4. Explain Shunyam Saamyasamuccaye for solving the following (a) $7(x+1)=8(x+1)$ (b) $12x-6x+3x+4x=0$.	CO2
9	Weekly Homework Assignments	3	Numbers, Ratio, Proportion, Variation, Averages, Percentages, Profit & loss, Time & Distance, Time & Work and related Shakuntala Devi Puzzles	1. The Bus Number: While visiting a small town in the United States, I lost my overcoat in a bus. When I reported the matter to the bus company I was asked the number of the bus. Though I did not remember the exact number I did remember that the bus number had a certain peculiarity about it. The number plate showed the bus number as a perfect square and also if the plate was turned upside down.? the number would still be a perfect square. I came to know from the bus company they had only five hundred buses numbered from 1 to 500. 2. Solve the puzzles based on the concept of Variation, Averages and Percentages. Farmer Thimmayya bought some mules at Rs. 50 each, sheep at Rs. 40 each, goats at Rs. 25 each, and pigs at Rs. 10 each. The average price of the animals per head worked	CO3

				to Rs. 30. What is the minimum number of animals of each kind did he buy? 3. Solve the puzzles based on the concept of Time and Distance. Two trains start towards each other from two stations 50 miles apart, at the same time and on a single track. Just when the trains' start out, a falcon leaves the first train and flies directly to the other train, and as soon as it reaches the second train, the bird starts back towards the first train. It continues doing so, flying backwards and forwards from one train to the other until the trains meet. Both the trains travel at a speed of 25 miles per hour, and the bird flies at 100 miles per hour. How many miles will the falcon have flown before the trains met?	
13	Weekly Homework Assignments	4	Clocks, Calendars, Cubes, logical connectives, Deductions, Linear and circular arrangements, Number and letter series, Coding and Decoding, Binary Logic.	1. A clock rings at 2 O'clock, 3 O'clock and so on. Then calculate the number of rings in a week. 2. A cube is cut parallel to one face by making 10 cuts such that all the resulting pieces are identical. What is the maximum number of identical pieces that can be obtained now by making 2 more cuts in any direction? 3. Consider the following logical statements: (1) If John is a student, then he has a library card. ($S \rightarrow L$) (2) If John has a library card, then he can borrow books. ($L \rightarrow B$) (3) John does not have a library card. ($\neg L$). Determine whether John can borrow books and whether John is a student. 4. In a country, there are three categories of persons Truth tellers, liars and alternators. Praveen met three persons Honey, Bunny, and Cherry from that country. Each of the three persons	CO4

				belongs to a different category. When asked about the details of the cities in the country, each of them made two statements. Honey : City X is 30 km to the North of City Y. I am a liar. Bunny : City W is 50 km away to the west of City Z. City M is 50 km away to the south of city W. Cherry : City Y is not to the west of city Z. Honey is a truth teller 1. If the distance between city y and city Z is 10 km, then how far is city W from city X? a. 40 km b. 50 km c. $30\sqrt{5}$ km d. $40\sqrt{5}$ km 2. Who among them is a truth teller? a. Honey b. Bunny c. Cherry d. Either (B) or (C)	
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COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	V-S21,V-S22,V-S23,V-S24	V-S21,V-S22,V-S23,V-S24	---	---	V-S9,V-S10,V-S11,V-S12	V-S9,V-S10,V-S11,V-S12	-	-	-
	Tutorial	--	--	---	---	--	--	-	-	-
	Lab	--	--	---	---	--	--	-	-	-
	Skilling	--	--	---	---	--	--	-	-	-
Tue	Theory	V-S37,V-S38,V-S39,V-S40	V-S37,V-S38,V-S39,V-S40	V-S49,V-S50,V-S51,V-S52	V-S49,V-S50,V-S51,V-S52	---	---	-	-	-
	Tutorial	--	--	--	--	---	---	-	-	-
	Lab	--	--	--	--	---	---	-	-	-
	Skilling	--	--	--	--	---	---	-	-	-

Wed	Theory	V-S41,V-S42,V-S43,V-S44	V-S41,V-S42,V-S43,V-S44	---	---	V-S25,V-S26,V-S27,V-S28	V-S25,V-S26,V-S27,V-S28	-	-	-
	Tutorial	--	--	---	---	--	--	-	-	-
	Lab	--	--	---	---	--	--	-	-	-
	Skilling	--	--	---	---	--	--	-	-	-
Thu	Theory	---	---	---	---	V-S1,V-S2,V-S3,V-S4,V-S5,V-S6,V-S7,V-S8	V-S1,V-S2,V-S3,V-S4,V-S5,V-S6,V-S7,V-S8	-	-	-
	Tutorial	---	---	---	---	--	--	-	-	-
	Lab	---	---	---	---	--	--	-	-	-
	Skilling	---	---	---	---	--	--	-	-	-
Fri	Theory	V-S33,V-S34,V-S35,V-S36	V-S33,V-S34,V-S35,V-S36	---	---	---	---	-	-	-
	Tutorial	--	--	---	---	---	---	-	-	-
	Lab	--	--	---	---	---	---	-	-	-
	Skilling	--	--	---	---	---	---	-	-	-
Sat	Theory	V-S17,V-S18,V-S19,V-S20,V-S45,V-S46,V-S47,V-S48	V-S17,V-S18,V-S19,V-S20,V-S45,V-S46,V-S47,V-S48	V-S13,V-S14,V-S15,V-S16	V-S13,V-S14,V-S15,V-S16	---	---	-	-	-
	Tutorial	--	--	--	--	---	---	-	-	-
	Lab	--	--	--	--	---	---	-	-	-
	Skilling	--	--	--	--	---	---	-	-	-

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Sun	Theory	--	--	--	--	--	--	-	-	-
	Tutorial	--	--	--	--	--	--	-	-	-
	Lab	--	--	--	--	--	--	-	-	-
	Skilling	--	--	--	--	--	--	-	-	-

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4
End Semester Summative Evaluation Total= 40 %	End Semester Exam (online MCQ)	Weightage	40		180	10	10	10	10
		Max Marks	100			25	25	25	25
In Semester Formative Evaluation Total= 25 %	Home Assignment and Textbook	Weightage	10		120	2.5	2.5	2.5	2.5
		Max Marks	100			25	25	25	25
	ALM	Weightage	15		120	3.75	3.75	3.75	3.75
		Max Marks	100			25	25	25	25
In Semester Summative Evaluation Total= 35 %	Semester in Exam-II (MCQ)	Weightage	17.5		90			8.75	8.75
		Max Marks	50					25	25
	Semester in Exam-I (MCQ)	Weightage	17.5		90	8.75	8.75		
		Max Marks	50			25	25		

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments.

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Satish Kumar Deevi	L	17-MA,12-MA	-	-	-	-
Ragamayi Sistla	L	48-MA,35-MA	-	-	-	-
Maddali Madhavi	L	2-MA,52-MA,41-MA,20-MA	-	-	-	-
Bhargavi Yella	L	39-MA,21-MA,14-MA	-	-	-	-
Bindu Pathuri	L	49-MA,25-MA,23-MA,19-MA,7-MA,10-MA	-	-	-	-
Ankita Tiwari	L	16-MA,18-MA,34-MA	-	-	-	-
Vijaya Nalleboyina	L	13-MA,45-MA,40-MA	-	-	-	-
S Ibrahim	L	22-MA,8-MA,11-MA	-	-	-	-
Imran Ali	L	27-MA	-	-	-	-
Sravani Jayanti	L	15-MA,24-MA,3-MA	-	-	-	-

LEELAVATHI REKAPALLI	L	33-MA,6- MA	-	-	-	-
Nirakara Pradhan	L	37-MA,9- MA,4- MA,42- MA,46- MA,51-MA	-	-	-	-
SEELAM DEVI	L	38-MA,5- MA,43- MA,28- MA,47-MA	-	-	-	-
SOMNATH KARMAKAR	L	1-MA,36- MA,50- MA,26- MA,44-MA	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Bhargavi Yella)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of Maths

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal) [object HTMLDivElement]